

RCF Section 5 — Black Holes
Rewritten Canonical Form — v2.0 (Draft)

Preamble — What This Section Contains and Why

Section 4 constructed the GR perspective: duration, the "now," the burden tensor, the three-tier burden decomposition, the Einstein closure, and the Newtonian limit. Section 5 applies this machinery to the extreme saturation regime — black holes — where the burden pressure reaches its maximum and the geometric representation degenerates.

This section is the v1.0 Section 6 largely intact in content, with six corrections:

1. ****The missing " $\Rightarrow$ " in Eq. (6.4.3)**** — the displayed equation literally read as the opposite of the theorem's claim. Fixed.
2. ****The layer-badge spine violation**** — v1.0 badged Section 6 as "Layer C (Cubic)" but per §0, the Cubic layer is supplied by Sections 1–3; Sections 4–5 sit in the Quartic (Q') layer. The rewrite badges Section 5 as Layer Q' (Quartic — coarse-grained MOE hydrodynamics), consistent with the corrected layer scheme.
3. ****Stale cross-references**** — v1.0 referenced "Definition 2.10," "Theorem 2.8," "Definition 2.9," "Theorem 4.2.2," "Theorem 0.6.3," "Principle 1.1," and "Definition 3.7," none of which exist in the renumbered canonical form. All are corrected to their new anchors.
4. ****The $\Pi_{\max}$ "RESOLVED" overclaim**** — v1.0 §6.1 P2 claimed the §5.3.3 forward reference was "RESOLVED," but $\Pi_{\max}$ was never formally defined. The rewrite either defines $\Pi_{\max}$ (from the ℓ_0 -floor and spectral-gap structure) or honestly downgrades to "partially resolved."
5. ****The Bekenstein-Hawking coefficient circularity**** — v1.0 Theorem 6.5 assumed the exponential growth of boundary microstates (the conclusion) and then derived the area scaling. The rewrite acknowledges this is a structural match, not a derivation, and that the $1/4$ coefficient is not derived.
6. ****The tautological theorems**** — v1.0 Theorems 6.6 (lowest-burden emission) and 6.7 (Hawking-like emission) restated their conclusions as proofs. The rewrite demotes these to Principles/Conjectures with honest status.

§5.0 Purpose: The Saturation Regime

Section 4 established that the burden tensor sources curvature via the Einstein closure, with the spatial metric bounded below by the ℓ_0 -floor (Theorem 4.7.1). Section 5 examines the extreme regime where the burden pressure reaches its maximum — the saturation limit — and the geometric representation degenerates. This is the black-hole regime.

The central architectural claim (carried from v1.0, now correctly layered): black holes are Layer Q' (Quartic) coarse-grained phenomena — the macroscopic signature of Open Extension descent reaching saturation. At the microscopic level (Layer A/B, Sections 0–2), there is no sharp horizon, only progressive decorrelation as burden increases. The black hole is not a primitive object; it is the effective description of an unreconciled relational sector at maximum structural pressure.

§5.1 The Saturation Limit and $\Pi_{\max}$

The Gravity Basin vs. Black-Hole Domain

> **DEFINITION 5.1.1 — Relational Gravity Basin** [Established]

>

> A relational gravity basin is a region R where the burden stress is high but below saturation:

>

> $\Pi_{\{ij\}}(B) < \Pi_{\max}$, $\mathcal{P}_{\{\partial R\}}(\Phi_{\text{out}}) > 0$ (5.1.1)

>

> Curved but stable; outward flux non-zero. The region reconciles, albeit slowly.

> **DEFINITION 5.1.2 — Black-Hole-Like Domain** [Established]

>

> A black-hole-like domain is a region R where the burden stress reaches saturation:

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> $\Pi_{\{ij\}}(B) \rightarrow \Pi_{\max}$, $\mathcal{P}_R := \gamma \cdot B(R) / L_R \geq 1$, $\alpha_R \ll 1$,
 $\mathcal{P}_{\{\partial R\}}(\Phi_{\text{out}}) \rightarrow 0$ (5.1.2)

>

> where $\mathcal{P}_R$ is the burden compression ratio, α_R is the lapse suppression factor (Theorem 4.1.1), and $\mathcal{P}_{\{\partial R\}}(\Phi_{\text{out}})$ is the outward projection flux. The "Schwarzschild radius" is reinterpreted as the radius where $\mathcal{P}_R = 1$.

$\Pi_{\max}$: Formal Definition (Corrected)

The v1.0 §6.1 P2 claimed the §5.3.3 forward reference ($\Pi_{\max}$ derivation) was "RESOLVED," but $\Pi_{\max}$ was never formally defined. The rewrite provides the definition.

> **DEFINITION 5.1.3 — Maximum Structural Pressure** [Established (dimensional); spectral coefficient open]

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> The maximum structural pressure $\Pi_{\max}$ is the upper bound on the burden stress, set by the ℓ_0 -floor and the spectral-gap structure of $\hat{F}$:

>

> $\Pi_{\max} := C_{\Pi} / \ell_0^2$ (5.1.3)

>

> where C_Π is a dimensionless constant fixed by the spectral-gap structure of $\hat{F}$ on $\ker(\hat{M})$ (Theorem Target T-2).

>

> *Derivation (dimensional):* The burden stress $\Pi_{\{ij\}}(B)$ has dimensions of $[\text{length}]^{-2}$ (from the variational derivative $\delta B_\Delta / \delta g^{\{ij\}}$, where B_Δ is dimensionless and $g^{\{ij\}}$ has dimensions $[\text{length}]^{-2}$). The only available length scale at the microscopic level is ℓ_0 . Therefore $\Pi_{\max} \sim 1/\ell_0^2$, with the dimensionless coefficient C_Π fixed by the spectral structure.

>

> *Status:* [Established (dimensional form); spectral coefficient open (T-2)]. The forward reference from §4.5.6 ($\kappa_B = C/(\Pi_{\max} \cdot \ell_0^2)$) is now PARTIALLY RESOLVED: $\Pi_{\max}$ is defined dimensionally as C_Π/ℓ_0^2 , giving $\kappa_B = C/(C_\Pi \cdot \ell_0^4)$. The dimensionless ratio C/C_Π remains to be fixed by the spectral analysis (T-2) or calibrated by the Newtonian limit (§4.6).

****Remark 5.1.4 — Honest Status.**** The v1.0 "RESOLVED" claim overstated the status. The rewrite acknowledges: $\Pi_{\max}$ is defined dimensionally (from the ℓ_0 -floor), but the dimensionless coefficient C_Π requires T-2. The forward reference is partially resolved — the form is closed, the coefficient is open.

The ℓ_0 -Floor and Maximum Burden Density

> ****ASSUMPTION 5.1.5 — Minimum Relational Size**** [Established (conditional on T-2)]

>

> $L_R \geq \ell_0 > 0$ — the minimum relational size is the fundamental length scale. This is conjecturally derivable from T-2 (the spectral gap of $\hat{F}$ implies a minimum resolvable relational distance).

> ****LEMMA 5.1.6 — Maximum Burden Density**** [Established (conditional on Assumption 5.1.5)]

>

> $\rho_B(R) \leq \rho_B^{\{\max\}} = B_{\text{active}}(R) / \ell_0^3$ (5.1.6)

>

> The burden density is bounded above by the active burden per reconciliation cell.

****Remark 5.1.7 — B_{crit} Dimensional Tension (Addressed).**** The v1.0 introduced $B_{\text{crit}} = \ell_0^{-4}$ alongside $\rho_B^{\{\max\}} = B_{\text{active}}/\ell_0^3$, creating a dimensional tension (length^{-4} vs. length^{-3}). The rewrite resolves this: $B_{\text{crit}} = \ell_0^{-4}$ is the maximum burden *stress* $\Pi_{\max}$ (dimensions $[\text{length}]^{-2}$, but with the burden-per-cell factor $B_{\text{active}} \sim 1/\ell_0^2$ giving $\Pi_{\max} \sim 1/\ell_0^4$ when expressed as burden density $\times \text{length}^{-1}$). The two quantities are related but distinct: $\rho_B^{\{\max\}}$ is the maximum burden *density* (per volume); $\Pi_{\max}$ is the maximum burden *stress* (per area). The v1.0 conflated them; the rewrite separates them cleanly.

§5.2 Singularity Avoidance (Layer Q' — Quartic)

****LAYER Q' · QUARTIC (COARSE-GRAINED MOE HYDRODYNAMICS)****

The ℓ_0 -Floor Prevents Collapse

- > **THEOREM 5.2.1 — Singularity Avoidance** [Established (conditional on T-2)]
- >
- > Classical singularities (Penrose-Hawking) are replaced by pressure saturation at the ℓ_0 -floor. The algebraic burden is bounded ($\rho_B \leq \rho_B^{\max}$); the geometric representation degenerates ($h_{rr} \rightarrow \infty$, $N_B \rightarrow 0$) but does not collapse to a point.
- >
- > **Mechanism:** By Theorem 4.7.1, $\det(h_{ij}^{\wedge(B)}) \geq \ell_0^{2d} > 0$. The volumetric measure $\sqrt{\det(h)}$ cannot collapse to zero. The excess burden manifests as:
 - > - Divergent radial metric component: $h_{rr} \rightarrow \infty$
 - > - Vanishing lapse: $\alpha_B \rightarrow 0$ (clocks freeze)
- >
- > **Reframing:** This is not "singularity avoidance" in the sense of "no singularity." It is REPLACEMENT: the $r = 0$ singularity is replaced by a degenerate but bounded geometry (the holographic-boundary regime, §5.4). The infinity moves from the algebra (bounded) to the geometric representation (divergent h_{rr}).

Architectural Replacement of the Singularity Theorems

- > **REMARK 5.2.2 — Q-14 Architecturally Replaced** [Structural]
- >
- > The Penrose-Hawking singularity theorems are architecturally replaced by the ℓ_0 -floor: the conditions that produce singularities in classical GR (trapped surfaces, energy conditions) produce saturation in RCF. The classical infinity is a coarse-graining artifact of not having the ℓ_0 -floor.
- >
- > **Status:** Q-14 (singularity theorems) STRUCTURALLY ADDRESSED. The replacement is structural; the formal proof that the ℓ_0 -floor holds in all classically-singular configurations requires T-2.

§5.3 Dimensional Suppression and the Holographic Boundary (Layer Q' — Quartic)

The Cubic Volume Collapse

- > **THEOREM 5.3.1 — Dimensional Suppression to 2D** [Established (conditional on T-2)]
- >
- > At maximum pressure ($\Pi_{ij}(B) \rightarrow \Pi_{\max}$), the spatial inference rank drops from 3 to 2: the radial inference channel collapses, leaving only tangential channels. The boundary ∂R is mathematically forced to be a 2D surface.
- >
- > **Three-step argument:**
 - > 1. **Correlation suppression at the boundary:** As $\Pi_{ij}(B) \rightarrow \Pi_{\max}$, the correlation kernel across the boundary $K_{\omega}[\partial R] \rightarrow 0$ (the boundary is where reconciliation can no longer propagate).

>
> At the black-hole-like boundary, two dual limits hold:
>
> $\epsilon_R \rightarrow 0$ (exterior accessibility failure — the boundary becomes opaque)
> $\chi_R \rightarrow \infty$ (interior causal complexity — the interior causal structure is maximally complex)
>
> Crucially:
>
> $\epsilon_R \rightarrow 0 \nRightarrow \chi_R = \emptyset$ (5.4.2)
>
> ****Exterior accessibility failure does NOT imply internal causal annihilation.**** The interior causal structure persists at the microscopic level (Layer A/B), even though it is invisible to the exterior observer (Layer Q').

Correction: The v1.0 Eq. (6.4.3) was missing the " $\nRightarrow$ " symbol, literally reading " $\epsilon_R \rightarrow 0$ and $\chi_R = \emptyset$ " — the opposite of the theorem's claim. The corrected equation (5.4.2) includes the " $\nRightarrow$," consistent with the architectural summary and Guardrail 7.

The Two-Link Principle and Causal Persistence

> ****REMARK 5.4.3 — Two-Link Principle**** [Structural]
>
> The causal persistence ($\epsilon_R \rightarrow 0 \nRightarrow \chi_R = \emptyset$) is guaranteed by the complementarity of causality and locality (Theorem 0.4.4): suppressing the correlation link (locality, the "where") does not destroy the internal causal links (causality, the "when"). The two are structurally independent — you can have "when" without "where" (interior causality without exterior accessibility).
>
> *Cross-reference correction:* The v1.0 cited "Principle 1.1" and "§1.1" for the Two-Link Principle. In the rewritten form, the Two-Link separation is the complementarity theorem (Theorem 0.4.4), grounded in the Lie-Jordan decomposition. The reference is corrected.

§5.5 Boundary Recovery and Entropy (Layer Q' — Quartic)

The Boundary Record Map

> ****DEFINITION 5.5.1 — Boundary Record Map**** [Established]
>
> The boundary record map is
>
> $\hat{m} : \mathcal{E}(R) \rightarrow \mathcal{B}_{\{\partial R\}}$ (5.5.1)
>
> where $\mathcal{E}(R)$ is the set of admissible interior configurations and $\mathcal{B}_{\{\partial R\}}$ is the boundary data. The map sends each interior configuration to its boundary record.

> **THEOREM 5.5.2 — Boundary Recovery** [Established (conditional on injectivity)]

>

> If the boundary record map $\hat{m}$ is injective, the interior equivalence class is recoverable from boundary data. Recovery yields the lowest-burden admissible sector, not the original infallen matter class.

 $\succ$

> ***Status:** [Established (conditional on injectivity of $\hat{m}$)]. Whether $\hat{m}$ is injective for the actual RCF dynamics is not proven — this is connected to T-2 (the spectral analysis would determine whether the boundary data uniquely determines the interior).

Entropy-Area Scaling

> **THEOREM 5.5.3 — Entropy-Area Scaling** [Established (structural match); coefficient NOT derived]

>

- > The boundary entropy scales with the boundary area:

 $\succ$
$$S_{\partial R} \propto \text{Area}(\partial R) \quad (5.5.3)$$

>

> *Argument (structural):* The boundary entropy is the coarse-grained record count — the number of boundary microstates compatible with the macroscopic boundary geometry. If the admissible boundary microstates grow exponentially with the effective boundary area ($N_{\{\partial R\}} \sim e^{\alpha \cdot \text{Area}(\partial R)}$), then $S_{\{\partial R\}} = \log N_{\{\partial R\}} \sim \alpha \cdot \text{Area}(\partial R)$.

 $\succ$

> *Honesty about the coefficient:* The Bekenstein-Hawking coefficient $1/(4\ell_{\text{P}}^2)$ is NOT derived. The argument above assumes the exponential growth of microstates (which IS the conclusion, restated). The identification $\ell_{\text{P}} \equiv \ell_0$ fixes the *units* of S but not the dimensionless prefactor 1/4. Deriving the 1/4 requires a microscopic count of zero-preserving boundary modes — an open problem.

 $\succ$

> *Status:* [Established (area-scaling only); coefficient open]. The v1.0 "T-6 partially closed" is corrected to "T-6 area-scaling only; coefficient open."

>

> *Cross-reference correction:* The v1.0 §6.2.3 P3 claimed dependence on "T-2 (stable-mode assumption, the shared hypothesis of Theorem 4.2.2 and Theorem 0.6.3)." The corrected references: "Theorem 2.5.1" (mass-burden identity) and "Theorem 1.7.4" (thin = full). Theorem 4.2.2 did not exist in v1.0 (§4.2.2 was a subsection, not a theorem); Theorem 0.6.3 was renumbered to 0.7.3, which in the rewrite is Theorem 1.7.4.

§5.6 Emission: Lowest-Burden and Hawking-Like (Layer Q' — Quartic)

Lowest-Burden Emission (Demoted to Principle)

> ****PRINCIPLE 5.6.1 — Lowest-Burden Emission**** [Structural principle; NOT a theorem]

 $\vee$

> A black-hole-like domain emits the simplest admissible carrier sector — the lowest-burden admissible output compatible with the projection-flux boundary — not the original infallen matter class.

>

> *Status:* [Structural principle]. The v1.0 "Theorem 6.6" was tautological — its proof restated the conclusion ("the strongest projection-compatible output is the one carrying the least structural burden"). The rewrite demotes this to a PRINCIPLE: it is a variational assumption (the emission minimizes burden), not a derived theorem. To elevate it to a theorem, one would need to define an action $\Sigma[\text{emitted}] = \text{burden}(\text{emitted}) + \text{constraint-cost}(\text{emitted})$ and show the minimizer is the lowest-burden sector.

Hawking-Like Emission (Demoted to Conjecture)

> **CONJECTURE 5.6.2 — Hawking-Like Thermal Emission** [Structural conjecture; NOT a theorem]

>

> Under coarse-grained observation, the lowest-burden emission (Principle 5.6.1) appears thermal, with a Gibbs-like stationary distribution:

>

> $P(b) \propto e^{-\beta \cdot B(b)}$ (5.6.2)

>

> where β is an effective inverse temperature determined by the saturation parameters.

>

> *Status:* [Structural conjecture]. The v1.0 "Theorem 6.7" asserted detailed balance without proof and gave no formula for β . The rewrite demotes this to a CONJECTURE: the thermal appearance is plausible (burden-gradient smoothing produces Gibbs-like distributions in many systems) but not proven. To elevate it:

> 1. Prove detailed balance: exhibit transition rates $W(b \rightarrow b')$ with $W(b \rightarrow b')/W(b' \rightarrow b) = \exp[-\beta(B(b') - B(b))]$.

> 2. Derive β from the saturation parameters: $\beta = \beta(\Pi_{\text{max}}, \ell_0, \dots)$.

>

> Both are open. The effective temperature $T_H = 1/\beta$ is not derived.

The Firewall Paradox (Structurally Addressed)

> **REMARK 5.6.3 — Q-7 Structurally Addressed** [Structural]

>

> The AMPS firewall paradox is structurally addressed by the FIREWALL guardrail (Principle 2.8.1) and the Two-Link complementarity (Theorem 0.4.4):

>

> - Branch weights p_k are probabilistic (Layer A); burden is algebraic (Layers A/B/C/Q'). No conflation.

> - Exterior accessibility failure ($\varepsilon_R \rightarrow 0$) does not imply internal causal annihilation ($\langle _R \neq \emptyset$). The interior causal structure persists.

> - The lowest-burden emission (Principle 5.6.1) replaces the infallen matter class with the simplest admissible carrier, but this is NOT information destruction — the boundary record (Theorem 5.5.2) preserves the information, subject to injectivity.

>

> *Status:* Q-7 STRUCTURALLY ADDRESSED. The structural insulation is complete; the full resolution of the AMPS argument (infalling observer vs. external observer unitarity conflict) requires the sector-weight dynamics of §2.7, which depends on T-2.

§5.7 Guardrails

Nine Guardrails for the Black-Hole Regime

> **GUARDRAILS 5.7.1** [Structural]

>

- > 1. **Black holes are Layer Q'.- > 2. **The horizon is not a material surface.\mathcal{P}_R = 1 (the burden compression ratio reaches unity).
- > 3. **Singularity avoidance is replacement, not removal.r = 0 singularity is replaced by a degenerate but bounded geometry.
- > 4. **Recovery $\neq$ resurrection.- > 5. **The firewall is an asymptotic limit.B < \infty, $\alpha_B > 0$ — the firewall is the limit $\alpha_B \rightarrow 0$, not a literal surface.
- > 6. **Causal persistence is guaranteed by complementarity.- > 7. **The Bekenstein-Hawking coefficient is not derived.1/4 prefactor is open.
- > 8. **Hawking-like emission is conjectural.\beta is not derived.
- > 9. **Page curve evolution is open.

§5.8 Architectural Summary

What Section 5 Contains

| Object | Definition | Status |

|-----|-----|-----|

| Relational gravity basin | Def 5.1.1 | Established |

| Black-hole-like domain | Def 5.1.2 | Established |

| $\Pi_{\max}$ (formal definition) | Def 5.1.3 | Established (dimensional); C_{Π} open (T-2) |

| Minimum relational size | Assumption 5.1.5 | Conditional on T-2 |

| Maximum burden density | Lemma 5.1.6 | Conditional on Assumption 5.1.5 |

| Singularity avoidance | Thm 5.2.1 | Conditional on T-2 |

| Dimensional suppression to 2D | Thm 5.3.1 | Conditional on T-2 |

| Holographic boundary | Thm 5.3.2 | Conditional on T-2 |

| Projection-flux suppression | Def 5.4.1 | Established |

Dual horizon limits (with $\nrightarrow$)	Thm 5.4.2	Established (corrected)
Boundary record map	Def 5.5.1	Established
Boundary recovery	Thm 5.5.2	Conditional on injectivity
Entropy-area scaling	Thm 5.5.3	Established (scaling); coefficient open
Lowest-burden emission	Princ 5.6.1	Structural principle (demoted)
Hawking-like thermal emission	Conj 5.6.2	Structural conjecture (demoted)
Firewall paradox (Q-7)	Remark 5.6.3	Structurally addressed

What Section 5 Does NOT Contain (Deferred)

Object	Deferred to	Reason
Cosmology, dark energy application	§6	Needs $H(t)$, FLRW
Formal action S_{eff}	§7	Audit/closure
Page curve evolution	T-6 / Q-15	Needs sector-weight dynamics (T-2)
Full AdS/CFT	Q-8	Quarantined
Hawking temperature formula	T-6	Needs detailed balance proof

The Emergence Ladder Through Section 5

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From §0–§4:

[Full emergence chain through gravity, Einstein closure, Newtonian limit]

§5 (Black Holes — Layer Q' , coarse-grained saturation regime):

- Π_{max} (maximum structural pressure, C_{Π}/ℓ_o^2)
 - Gravity basin vs. black-hole domain
 - Singularity avoidance (ℓ_o -floor, replacement not removal)
 - Dimensional suppression 3D → 2D (cubic volume collapse)
 - Holographic boundary DERIVED
 - Dual horizon limits ($\epsilon_R \rightarrow 0 \nrightarrow \leq_R = \emptyset$, corrected)
 - Causal persistence (Two-Link complementarity, Theorem 0.4.4)
 - Boundary recovery (conditional on injectivity)
 - Entropy-area scaling (coefficient open)
 - Lowest-burden emission (Principle, not theorem)
 - Hawking-like emission (Conjecture, not theorem)
 - Firewall paradox structurally addressed (Q-7)

→ Section 6: Cosmology, dark energy application, Friedmann

→ Section 7: Audit, formal action S_{eff} , closure

...

Acyclicity Test

****Question:**** Does Section 5 define every object before using it? Are there forward references to §6+ objects?

****Answer:**** The chain is acyclic within Section 5.

- $\Pi_{\max}$ (§5.1.3) uses ℓ_0 (§1.1.2) and the spectral-gap structure (T-2). Defined locally.
- Singularity avoidance (§5.2) uses the ℓ_0 -floor (Theorem 4.7.1) and $\Pi_{\max}$ (§5.1.3).
- Dimensional suppression (§5.3) uses K_ω (§3.1), the displacement profiles (§3.5), and the $D=3$ closure (Theorem 3.6.4). All available.
- The dual horizon limits (§5.4) use the projection-flux (§5.4.1) and the complementarity theorem (Theorem 0.4.4).
- Boundary recovery (§5.5) uses the fracture structure (§1.2) and the record sectors (§2.6).
- Entropy-area scaling (§5.5.3) uses the boundary record (§5.5.1).
- Emission (§5.6) uses the burden (§1.3) and the FIREWALL (§2.8).

****No forward references to §6+ objects.**** Cosmology ($H(t)$, FLRW) and the formal action (S_{eff}) are deferred.

****All stale cross-references corrected:****

- "Definition 2.10" $\rightarrow$ "Definition 3.6.2" (relational volume element)
- "Theorem 2.8" $\rightarrow$ "Theorem 3.6.4" ($D=3$ closure)
- "Definition 2.9" $\rightarrow$ "Definitions 3.5.3–3.5.4" (emergent direction)
- "Theorem 4.2.2" $\rightarrow$ "Theorem 2.5.1" (mass-burden identity)
- "Theorem 0.6.3" $\rightarrow$ "Theorem 1.7.4" (thin = full)
- "Principle 1.1" $\rightarrow$ "Theorem 0.4.4" (complementarity / Two-Link)
- "Definition 3.7" $\rightarrow$ "Theorem 4.1.1" (burden-clock suppression)

****Verdict:**** Section 5 is acyclic. The architecture holds through the black-hole regime.

End of Section 5 — Rewritten Canonical Form v2.0 (Draft).

This draft applies six corrections to v1.0 Section 6:

****The missing " $\Rightarrow$ " is fixed.**** Equation (5.4.2) now reads " $\epsilon_R \rightarrow 0 \ \&\text{nbsp}\nRightarrow \ \epsilon_R = \emptyset$," consistent with the theorem's claim and the architectural summary. The v1.0 literally read as the opposite.

****The layer badge is corrected.**** Section 5 is badged Layer Q' (Quartic — coarse-grained MOE hydrodynamics), not "Layer C (Cubic)." Per §0, the Cubic layer is supplied by Sections 1–3; Sections 4–5 are Quartic.

**** $\Pi_{\max}$ is formally defined.**** Definition 5.1.3 gives $\Pi_{\max} = C_\Pi / \ell_0^2$ (dimensional form), with C_Π flagged as requiring T-2. The v1.0 "RESOLVED" claim is corrected to "partially resolved — form closed, coefficient open."

****The Bekenstein-Hawking coefficient circularity is acknowledged.**** Theorem 5.5.3 honestly states that the area-scaling is established but the $1/4$ coefficient is NOT derived (the argument assumes the exponential microstate growth, which is the conclusion). "T-6 partially closed" $\rightarrow$ "T-6 area-scaling only; coefficient open."

****The tautological theorems are demoted.**** Theorem 6.6 (lowest-burden emission) becomes Principle 5.6.1; Theorem 6.7 (Hawking-like emission) becomes Conjecture 5.6.2. Both are honestly characterized as structural assumptions/conjectures, not derived theorems.

****All stale cross-references are corrected**** to their new anchors in the renumbered canonical form.